



Designing Bicycle Lane, Carl-Von-Ossiezký-Strasse, Chemnitz.



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Introduction

During the summer term of 2025, a group of dedicated students in Road Traffic Engineering at the University of Applied Sciences Zwickau undertook a design project titled 'Urban Traffic Facilities Design Project.' Their task was to develop an environmentally friendly and practical solution to enhance traffic safety and mobility along a particular section of road in the City of Chemnitz. The task was assigned by the Chemnitz administration to students of the University of Applied Sciences Zwickau, who were guided and supervised by professors from the University.

The study focused on the corridor on Carl-Von-Ossietzky-Strasse between Claus-Strasse and Gablenz, where the most pressing issue was the lack of a properly designed infrastructure for cyclists. This absence created challenges for both cyclists and motorists, as the roadway space was unsegregated.

The project aimed to analyze the interaction between cyclists and vehicles, exploring ways to make it safer and more efficient. Through collaboration with university staff and the city government, the team developed a proposal aligned with modern urban mobility principles, contributing to Chemnitz's future infrastructure development.

Motivation:

Carl-von-Ossietzky-Straße is a prime candidate for environmentally friendly and space-saving mobility through the implementation of an advisory cycle lane. Due to the road's narrowness and mixed traffic conditions, a completely segregated cycle track would be both impractical and costly. An advisory cycle lane offers a tangible, low-cost solution that requires no major infrastructural restructuring — just simple road markings. This measure would enhance cycling safety by reducing motorized traffic speeds, especially where traffic volumes fluctuate. By encouraging cycling with such a modest intervention, the road can benefit from reduced congestion, lower emissions, and improved overall mobility. Therefore, a cycle advisory lane is the most suitable and effective option for Carl-von-Ossietzky-Straße, in line with the city's policy for environmentally sustainable urban transport.



Tasks

The students were asked to design a solution for cycle routes on Carl von Ossietzky Straße. Currently, there are no cycle routes along Carl-von-Ossietzky-Straße, except for the crossing with the "NUMICO" route on the Hans-Ziegler-Straße axis. The total length for which we were asked to design the cycling infrastructure is approximately 1.7 km. Below, we have shared our proposed solution, along with the changes suggested by our group. The problem has been addressed by ensuring that all relevant rules and regulations were carefully considered in the development of our proposal.

Proposed Solution: Advisory Lanes

An advisory lane is a type of road marking used to guide cyclists while allowing flexibility for motorists, especially on narrower roads where a dedicated bike lane is not feasible. Unlike fully separated bike lanes, advisory lanes are indicated by dashed lines and help define space for cyclists without restricting vehicle movement.

This approach improves road safety by encouraging a structured interaction between cyclists and motorists while maintaining efficient traffic flow. In locations where space constraints prevent complete segregation, advisory lanes offer a practical compromise, promoting safer shared road use while acknowledging varying traffic demands.

While designing, it was kept in accordance with the RASSt that stipulates:

- **Standard width:** 1.5 meters
- **Minimum width (in constrained areas):** 1.25 meters

Key Benefits of Advisory Lanes

- Optimizes the use of available highway space without requiring full road reconstruction.
 - Cheaper than dedicated cycling paths.
 - Enhances the safety and mobility of cyclists.
 - Provides pedestrian footways and side planting.
 - Cheaper and better to maintain than fully separated facilities.
- Our system will be an efficient, safe, and sustainable transport solution that is compatible with national standards.

Current Scenario

Carl-Von-Ossietzky-Straße is one of the main roads that connects Luther-Straße with Südring, which is used heavily by motorized traffic, bicycle traffic as well as pedestrians. Particularly the patch of road which was given as a project, the area between Claus-Straße and Gablenz is considered very important. Residential buildings surround this 1.7-1.8 km road; it has a Car Service Centre, Gymnastics Club, Primary School, and a shopping complex.



Methodology

To accommodate the variations in road geometry and surrounding infrastructure, the target stretch was divided into five sections based on length and width. Each section was analyzed independently for optimal design adaptability..

Overview of Sections

Section	From – To	Length (m)	Width (m)
1	Clausstraße – Heimgartenstraße	244	20.3
2	Heimgartenstraße – Geibelstraße	716	23.5
3	Geibelstraße – Hans-Ziegler-Straße	261	18.2
4	Hans-Ziegler-Straße – Carl-von-Ossietzky	264	13.5
5	Carl-von-Ossietzky – Gablenz	292	13.7

Each section was assessed for:

- Roadway capacity
- Potential for advisory lane integration
- Pedestrian and vehicular interaction
- Space for green elements and parking

Design Considerations and Advantages

General Design Features (All Sections)

- **Advisory lane** on both sides for cyclists
- **Two vehicle lanes** (one in each direction)
- **Pedestrian sidewalks** on both sides
- **Green planting strips** where space allows
- **Central median strip** for safety and greenery (in wider sections)

Parking Provision

From Sections 1–3, which feature wider roadways, parallel parking lanes are incorporated, with a width of 2.5 meters, in accordance with RASSt guidelines. This measurement includes a 0.5-meter safety buffer to prevent dooring accidents, ensuring safer conditions for all road users.

Overall Advantages

- Promotes safe multimodal travel
- Maintains smooth vehicular flow
- Low-cost implementation with high impact
- Enhances the urban environment and visual appeal
- Complies with German urban planning and mobility guidelines

Detailed Section Analysis

Section 1: Clausstrasse – Heimgartenstrasse

- **Length:** 244 meters | **Width:** 20.3 meters



Design Changes: It has space for advisory lanes, parking lots, pedestrian paths, and a green center median. The design allows for all the users to coexist peacefully and adds quality to the urban street landscape. Previously, there was parking on both sides of the streets, but according to the proposed solution, we suggest removing parking from the left side, which is from the east of Claus-Strasse to Heimgartenstrasse, as the road width is narrower. With this proposed change, we are now able to incorporate advisory lanes on both sides while keeping all the necessary guidelines in consideration.

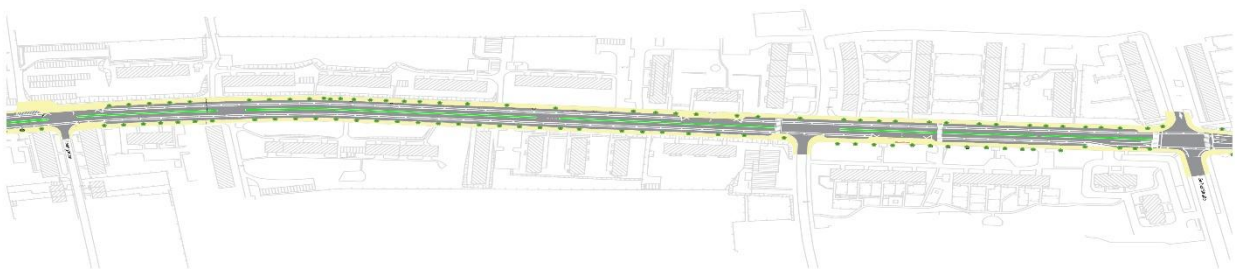


Section 2: Heimgartenstrasse – Geibelstrasse

- **Length:** 716 meters | **Width:** 23.5 meters



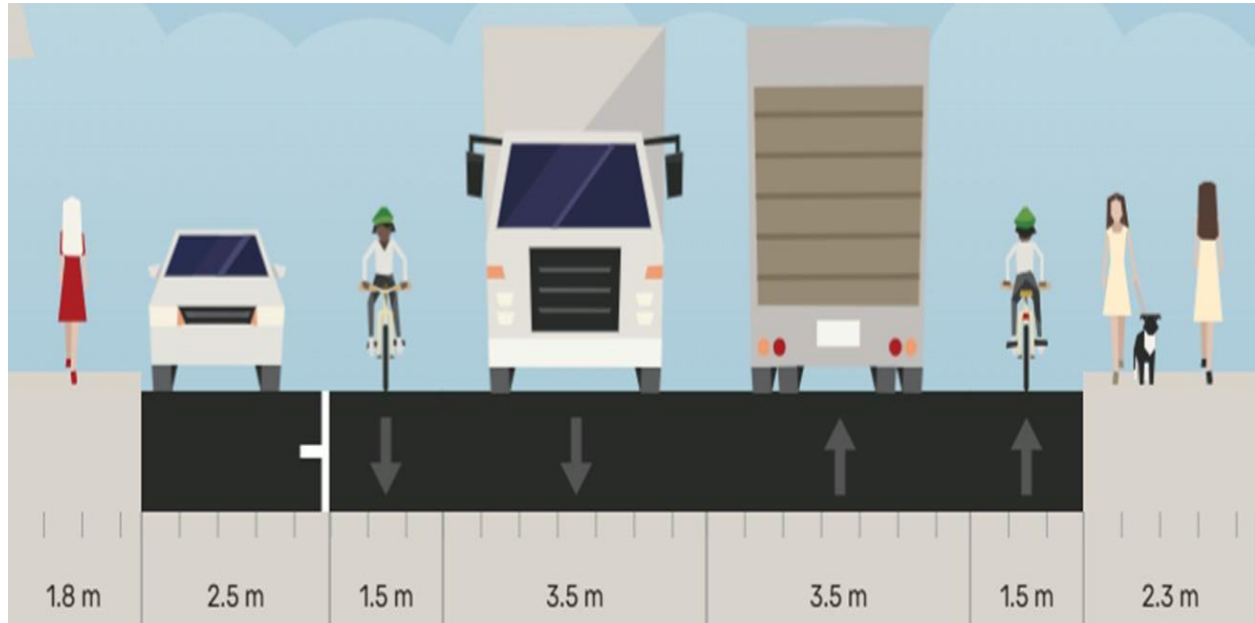
Design Changes: Between Heimgartenstrasse and Geibelstrasse, the road width expands from 20.3 meters to 23.5 meters, gaining an additional 3.2 meters. This extra space allows for the reinstatement of a parallel parking lane that was previously removed due to spatial limitations. Additionally, pedestrian areas are expanded, and the dividing line width increases slightly from 1.5 meters to 1.7 meters, making use of the available space.



Section 3: Geibelstrasse – Hans-Ziegler-Strasse

- **Length:** 261 meters | **Width:** 18.2 meters

The following street has been divided into two parts for better understanding as there are changes in the width of the road.

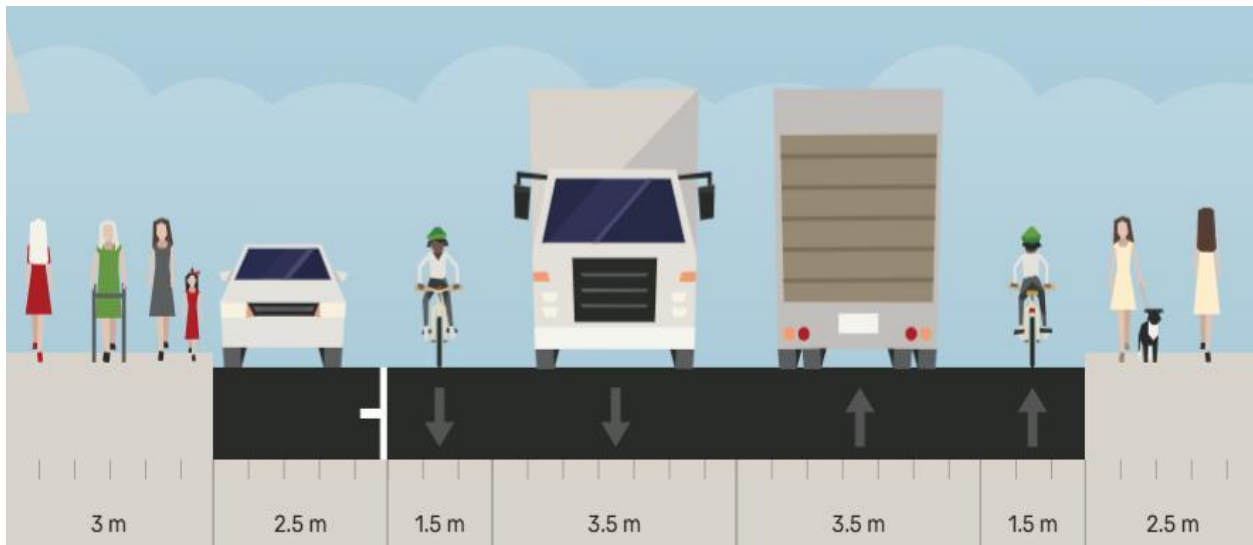


Part - 1 Geibelstrasse to Ernst-Enge-Strasse

Design Changes: Between Geibelstrasse and Ernst-Enge-Strasse, we have removed the island that was there for pedestrian crossing for better space for an advisory lane. We also suggest removing a tree that is in front of the "Hypnose Oelsner" to better accommodate temporary drop and pickup points as well as to better accommodate the advisory lanes. We also suggest removing 14 meters of parking from the east of Geibelstrasse towards Hans-Ziegler-Strasse to better accommodate space for pedestrians as well as an advisory lane.



Part - 2 Ernst-Enge-Strasse - Hans-Ziegler-Strasse

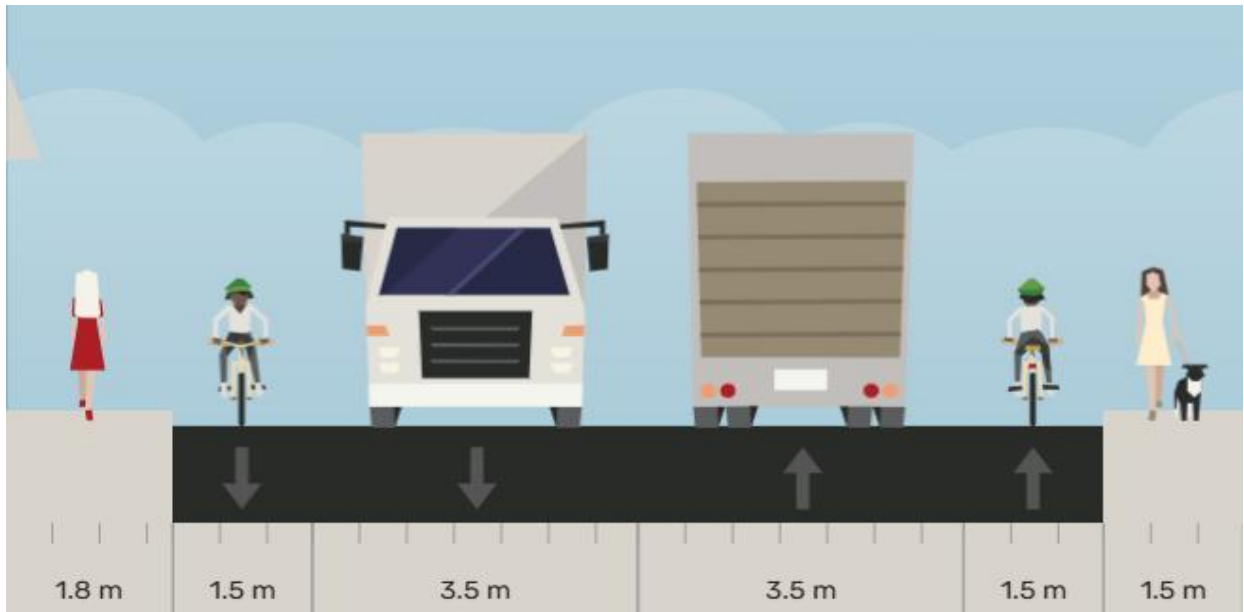


Design Changes: Between Ernst-Enge-Strasse - Hans-Ziegler-Strasse, in front of Gablenz Centre, we suggest changing diagonal parking to parallel parking to better accommodate the advisory lane. We had to resort to this solution due to the limited availability of width. We also suggest removing the advertising pillar in front of the Gablenz Centre to better accommodate pedestrian lanes.



Section 4: Hans-Ziegler-Strasse – Carl-von-Ossietzky-Strasse

- **Length:** 264 meters | **Width:** 13.5 meters

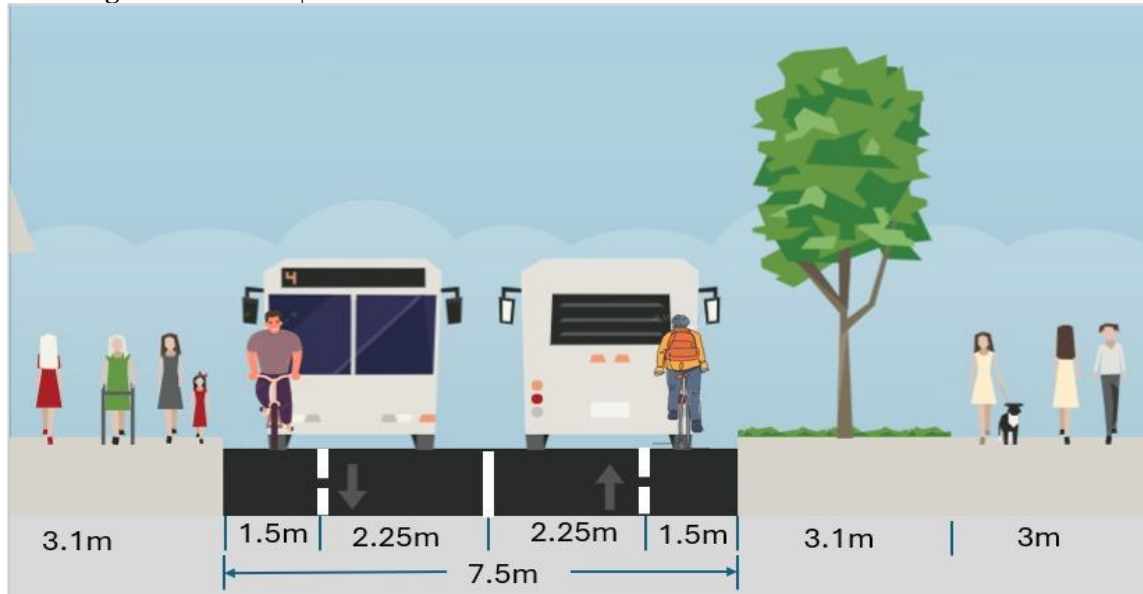


Design Changes: Previously, there was parking for taxis as well as for visiting guests and residents of the area, which has to be removed to better accommodate a separate advisory lane and ensure there is no conflict between motorized traffic and cyclists and to avoid reducing the space of pedestrians.

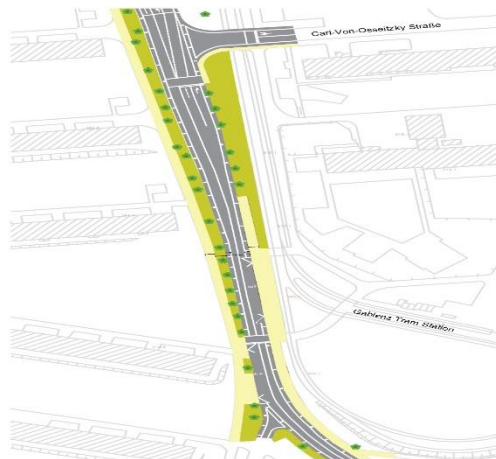


Section 5: Carl-von-Ossietzky-Strasse – Gablenz

- **Length:** 292 meters | **Width:** 13.7 meters



Design Changes: From Carl-von-Ossietzky-Strasse towards Gablenz, the road gets narrower. Before



the signalized intersection, the width of the road for vehicular traffic, as well as the advisory lane, is 10 meters, and it reduces to 7.5 meters just after 65 meters from the signalized intersection. The road narrows gradually, and a road sign will warn motorized vehicles to reduce their speed and share the road with cyclists.



References

- Street cross-sections created using Streetmix (<https://streetmix.net>).
- Dimension specifications based on AutoCAD plans provided by Prof. Felix Rudolph, WHZ.
- Maps and visuals obtained from Google Maps.
- Cycling advisory lane reference: German Road Safety Campaign (2022). 'Cycling in Germany'.
- Retrieved from:
https://www.germanroadsafety.de/fileadmin/user_upload/Broschueren/Fahrrad_fahren_in_Deutschland/Cycling-in-G